

Project 3 – Cloud Infrastructure

Case Study: Cloud Infrastructure Setup for a Growing Startup

A rapidly growing startup, TECH-Start, has decided to migrate their services to the AWS cloud to leverage its scalability and cost-effectiveness. They have recently hired you as a cloud administrator, to manage this transition and maintain their cloud infrastructure.

TECH-Start has a diverse team, including developers, data scientists, and marketing professionals, all of whom require different levels of access to the AWS resources. Your first task is to create multiple user accounts using AWS Identity and Access Management (IAM). Once the user accounts are set up, you need to define the permissions for each account. The developers need full access to the EC2 instances, while the data scientists require access to both EC2 and S3 storage. The marketing team, on the other hand, only needs read access to the S3 storage to analyze customer data.

Next, you will need to create an EC2 instance that will host TECH-Start web application and finally explore different storage options provided by AWS. You will create an S3 bucket and set up the necessary configurations.

This case study demonstrates how a cloud administrator can manage a company's AWS resources effectively, ensuring that each team member has the appropriate access levels and that the company's services run smoothly on the cloud.

Deliverables:

In this project, you need to connect on <https://aws.amazon.com/console/> and signup to create AWS free Tier account to build the environment for us. It includes the following tasks that should be completed by students.

- Create Multiple Users Accounts to define IAM concepts. Create groups.
- Set **permissions** on User Accounts.
- Create EC2 Instance.

- Understand different types of storage options and build / create S3 Storage.

Knowledge Checks:

1. Explain the importance of Availability Zones? In your response, please include the following:

- The definition of an Availability Zone.
- How Availability Zones contribute to the reliability and scalability of AWS services.

An example of how you might use Availability Zones in a real-world application.

Availability zones or AZ's are extremely crucial in AWS infrastructure.

Availability zones help with increased availability, fault tolerance, and higher scalability and reliability. Availability Zones are physically separate datacenters within an AWS Region. Each availability zone has independent power including, networking, and cooling, but it's connected to other AZs in the same region through low-latency, high-throughput, and redundant networking. Spreading applications across multiple availability zones protects from datacenter failures and fault tolerance. If one AZ experiences a power failure or a natural disaster, the client's application can continue operating from another AZ. AWS services can be deployed across multiple availability zones, which ensures the application remains available even during outages in one zone. Multiple AZs allow horizontal scaling across zones; For example, EC2 Auto Scaling can add instances in different AZs based on traffic, providing better performance and distribution.

For example, in a real world application such as running an e-commerce website using AWS, I could deploy my web servers, EC2 instances, in two AZs behind an Elastic Load Balancer. My database (Amazon RDS) is configured for Multi-Availability zone deployment, where the primary database runs in one AZ and the standby datacenter in another. If AZ A goes down, the load balancer automatically routes traffic to AZ B replica, and the RDS service promotes the standby database as the new main setup. Which ensures that customers can continue purchasing even if one datacenter fails.

2. Can you explain the differences and use cases of AWS CloudWatch and AWS CloudTrail and how you might use both AWS CloudWatch and AWS CloudTrail in a real-world application.?

Although they serve different purposes, AWS CloudWatch and AWS CloudTrail are both essential in monitoring and managing the AWS environment. AWS CloudWatch focuses on monitoring and performance because it provides metrics, logs, and alarms. CloudWatch allows tracking of health and performance of AWS resources such as EC2 instances, Lambda functions, and S3 buckets, and it also sets up automatic alerts based on predefined thresholds. AWS CloudTrail is focused and geared towards security and auditing by logging every API call made within an AWS account. CloudTrail records and monitors security and auditing by logging every API call made within an AWS account; Such as, who made the request, what resources were affected, and the outcome of the operation configured. In essence, CloudWatch ensures the reliability of the system while CloudTrail is useful for monitoring and tracking user activity, compliance, and investigating potential security breaches.

In a real world application, CloudWatch and CloudTrail can be used together to ensure the performance, security, and compliance of the AWS infrastructure. For example, if I'm managing an EC2 instance, CloudWatch would be used to monitor the health of the EC2 instance by tracking the metrics such as CPU utilization, memory usage, etc. If the EC2 instance CPU usage exceeds a certain threshold, CloudWatch can trigger an alarm to notify me, then I can investigate and take corrective action to avoid downtime. CloudTrail would track and log all the API calls and actions performed on the EC2 instance. For example, if I stop or start an instance, CloudTrail records that action with details about who initiated the change, what instance was affected, and when the change occurred. The information the CloudTrail provides is crucial for security, monitoring, and auditing purposes, because it helps identify any unauthorized or unexpected changes.

3. What is the importance of a well architected framework and how cost, security, performance efficiency, and operational excellence factor into a successful migration strategy.?

A well architected framework is exceedingly crucial when designing and migrating systems to the cloud, because it provides a set of practices that ensures the architecture is efficient, secure, monitored, and cost-effective. It helps clients and businesses to build scalable and sustainable cloud environments. AWS defines its well architected framework around cost, security, performance efficiency, and operational excellence. Each of these plays a key role in a successful migration strategy. Cost optimization helps to minimize expenses. During migration, cost helps to ensure the right balance between cost and performance, while reducing unnecessary costly expenses. Security is crucial for protecting sensitive data and ensuring compliance with regulations. During migration, security practices must be implemented starting with the data transfer to the configuration of cloud resources because it ensures that security risks are minimized, and important data is protected. Performance efficiency ensures that performance remains optimal while minimizing waste. Performance efficiency in the cloud prevents downtime and is important to continually assess and improve the performance of cloud resources, which allows for rapid scaling and agility. Operational excellence focuses on monitoring and improving processes and systems that keep the application in the cloud environment running smoothly. A successful migration strategy that includes all these characteristics ensures long term sustainability and optimal performance with best cost and security practices.

Submission:

Make sure to capture a screenshot after completing all the mentioned tasks. Upload the screenshots on the word document with the task name as a heading. Also answer all 3 knowledge check questions listed. Once the document is ready, upload it under the Project Submission section of your dashboard.

Creating User groups with permissions:

1. **Developers:**

e user group

Create user group

Name the group

User group name
Enter a meaningful name to identify this group.

Maximum 128 characters. Use alphanumeric and '+=,._-' characters.

Add users to the group - Optional (2)

Attach permissions policies - Optional (1/1045) [Info](#)

You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.

[X](#)

Filter by Type
All types ▼ 1 match

☒	Policy name	Type	Used as	Description
☒	 AmazonEC2FullAccess	AWS managed	Permissions policy (2)	Provides full access to Amazon EC2 via...

2. Data Scientists

Create user group

Name the group

User group name
Enter a meaningful name to identify this group.

Maximum 128 characters. Use alphanumeric and '+=,._-' characters.

3. Marketing

Create user group

Name the group

User group name

Enter a meaningful name to identify this group.

Maximum 128 characters. Use alphanumeric and '+=, @-_' characters.

Attach permissions policies - *Optional* (2/1045) [Info](#)

You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.

Filter by Type			
amazons3re	X	All types	1 match
☒	Policy name	Type	Used as
☒	AmazonS3ReadOnlyAccess	AWS managed	Permissions policy (1)

[Cancel](#)[Create user group](#)

Creating User Groups with Permissions:

aws

[Alt+S]

IAM > User groups

Identity and Access Management (IAM)

Dashboard

▼ Access management

User groups

Users

Roles

Policies

Identity providers

User groups (3) [Info](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

☐	Group name	Users	Permissions
☐	DataScientists	1	Defined
☐	DEVELOPERS	1	Defined
☐	MARKETING	1	Defined

1. Creating User in data science:

- Step 1
Specify user details
- Step 2
Set permissions
- Step 3
Review and create

Specify user details

User details

User name

data1

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☐ Provide user access to the AWS Management Console - optional

If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel

Next

- Step 1
Specify user details
- Step 2
Set permissions
- Step 3
Review and create

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☒ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1/3)

Create group

Search

< 1 > ⚙

☒	Group name	Users	Attached policies	Created
☒	DataScientists	1	AmazonEC2FullAccess and Amazo...	2025-04-21 (5 days ago)
☐	DEVELOPERS	1	AmazonEC2FullAccess	2025-04-21 (5 days ago)
☐	MARKETING	1	AmazonS3ReadOnlyAccess and A...	2025-04-21 (5 days ago)

IAM > Users > Create user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name

data2

Console password type

None

Require password reset

No

Permissions summary

Name

▲

Type

▼

Used as

▼

DataScientists

Group

Permissions group

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

2. Creating User in Developers:

IAM > Users > Create user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Specify user details

User details

User name

dev2

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☐ Provide user access to the AWS Management Console - optional

If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

📘 If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel

Next

aws [Search] [Alt+S] Global Cindy

IAM > Users > Create user

Step 1: Specify user details
Step 2: **Set permissions**
Step 3: Review and create

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

- ☒ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.
- ☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.
- ☐ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1/3) [Create group](#)

Search

☐	Group name	Users	Attached policies	Created
☐	DataScientists	1	AmazonEC2FullAccess and Amazo...	2025-04-21 (5 days ago)
☒	DEVELOPERS	1	AmazonEC2FullAccess	2025-04-21 (5 days ago)
☐	MARKETING	1	AmazonS3ReadOnlyAccess and A...	2025-04-21 (5 days ago)

► **Set permissions boundary - optional**

aws [Search] [Alt+S] Global Cindy

IAM > Users > Create user

Step 1: Specify user details
Step 2: Set permissions
Step 3: **Review and create**

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name dev2	Console password type None	Require password reset No
-------------------	-------------------------------	------------------------------

Permissions summary

Name	Type	Used as
DEVELOPERS	Group	Permissions group

Tags - optional
Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.
No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

3. Creating user in Marketing:

- Step 1
● **Specify user details**
- Step 2
○ Set permissions
- Step 3
○ Review and create

Specify user details

User details

User name

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

- ☐ Provide user access to the AWS Management Console - *optional*
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

📘 If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel Next

- Specify user details
- Step 2
● **Set permissions**
- Step 3
○ Review and create

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

- ☒ **Add user to group**
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User groups (1/3)

🔄 Create group

Group name	Users	Attached policies	Created
☐ DataScientists	1	AmazonEC2FullAccess and Amazo...	2025-04-21 (5 days ago)
☐ DEVELOPERS	1	AmazonEC2FullAccess	2025-04-21 (5 days ago)
☒ MARKETING	1	AmazonS3ReadOnlyAccess and A...	2025-04-21 (5 days ago)

▶ Set permissions boundary - *optional*

[illegible]

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management New

▼ Access reports

Access Analyzer

External access

Unused access

Analyzer settings

DataScientists Info

Summary

User group name

DataScientists

Creation time

April 21, 2025, 20:02 (UTC-05:00)

Users (1)

Permissions

Access Advisor

Permissions policies (2) Info

You can attach up to 10 managed policies.

Search

Filter by Type

All types

☐	Policy name 🔗	Type
☐	+ AmazonEC2FullAccess	AWS managed
☐	+ AmazonS3FullAccess	AWS managed

aws

Search

[Alt+S]

Global

IAM > Users > data1

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Access reports

Access Analyzer

External access

Unused access

Analyzer settings

data1

Info

Summary

ARN

arn:aws:iam::092896355827:user/data1

Console access

Disabled

Access key 1

Create access key

Created

April 21, 2025, 20:07 (UTC-05:00)

Last console sign-in

-

Permissions

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (2)

Remove

Add permission

Permissions are defined by policies attached to the user directly or through groups.

Search

Filter by Type

All types

< 1 >

	Policy name	Type	Attached via
☐	 AmazonEC2FullAccess	AWS managed	Group DataScientists
☐	 AmazonS3FullAccess	AWS managed	Group DataScientists

Developers:

Identity and Access Management (IAM) <

Dashboard

▼ Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management [New](#)

▼ Access reports

Access Analyzer

External access

DEVELOPERS [Info](#)

Summary

User group name
DEVELOPERS

Creation time
April 21, 2025, 20:01 (UTC-05:00)

Users
(1)

Permissions

Access Advisor

Permissions policies (1) [Info](#)

You can attach up to 10 managed policies.

Filter by Type

All types



Policy name [↗](#)



Type



AmazonEC2FullAccess

AWS managed

aws

Search

[Alt+S]

Global

IAM > Users > DEV1

DEV1

Info

Delete

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Access reports

Access Analyzer

External access

Unused access

Analyzer settings

Summary

ARN
arn:aws:iam::092896355827:user/DEV1

Console access
Disabled

Access key 1
Create access key

Created
April 21, 2025, 20:06 (UTC-05:00)

Last console sign-in
-

Permissions

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (1)

Remove

Add permissions

Permissions are defined by policies attached to the user directly or through groups.

Search

Filter by Type

All types

< 1 >

☐

Policy name

▲

Type

▼

Attached via

☐

AmazonEC2FullAccess

AWS managed

Group DEVELOPERS

Marketing:

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management New

Access reports

Access Analyzer

External access

Unused access

Analyzer settings

MARKETING Info

Summary

User group name
MARKETING

Creation time
April 21, 2025, 20:03 (UTC-05:00)

Users
(1)

Permissions

Access Advisor

Permissions policies (2) Info

You can attach up to 10 managed policies.

Filter by Type

Search

All types



Policy name



Type



AmazonS3ReadOnlyAccess

AWS managed



AmazonS3TablesReadOnlyAccess

AWS managed

aws

Search

[Alt+S]

Global

Cindy

IAM > Users > mark1

1

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Access reports

Access Analyzer

External access

Unused access

Analyzer settings

Credential report

mark1

Info

Delete

Summary

ARN
arn:aws:iam::092896355827:user/mark1

Console access
Disabled

Access key 1
Create access key

Created
April 21, 2025, 20:07 (UTC-05:00)

Last console sign-in
-

Permissions

Groups
(1)

Tags

Security credentials

Last Accessed

Permissions policies (2)

Permissions are defined by policies attached to the user directly or through groups.

Search

Filter by Type
All types

< 1 >

☐

Policy name

Type

Attached via

☐

AmazonS3ReadOnlyAccess

AWS managed

Group [MARKETING](#)

☐

AmazonS3TablesReadOnlyAccess

AWS managed

Group [MARKETING](#)

Users:



[IAM](#) > Users

Identity and Access Management (IAM) <

Dashboard

▼ Access management

User groups

Users

Roles

Policies

Identity providers

Users (3) [Info](#)

An IAM user is an identity with long-term credential

☐	User name	▲	Pair
☐	<u>data1</u>		/
☐	<u>DEV1</u>		/
☐	<u>mark1</u>		/

Creating and Launching an EC2 instance:

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

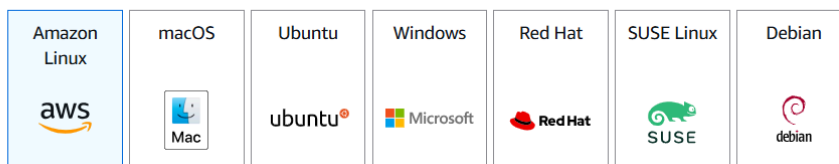
[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents

[Quick Start](#)



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI

Free tier eligible

▼

ami-060a84cbcb5c14844 (64-bit (x86), uefi-preferred) / ami-075adf72623183922 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.7.20250414.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	ⓘ	
64-bit (x86) ▼	uefi-preferred	ami-060a84cbcb5c14844	2025-04-11	ec2-user		Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Free tier eligible

▼

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software



EC2 > Instances > Launch an instance

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

[Create new key pair](#)

▼ Network settings [Info](#)

[Edit](#)

Network | [Info](#)

vpc-08d7819c567a82cfe

Subnet | [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group☐ Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

aws

Search

[Alt+S]

EC2 > Instances > Launch an instance

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere
0.0.0.0/0

☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

▼ Configure storage

Info

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

Launch instance

EC2 instance configured:

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) | All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
TECH Start	i-020a7650200bed52f	Running	t2.micro	2/2 checks passed	View alarms +	us-east-2a	ec2-3-129-26-249.us-east-2.compute.amazonaws.com

i-020a7650200bed52f (TECH Start)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID i-020a7650200bed52f	Public IPv4 address 3.129.26.249 open address	Private IPv4 addresses 172.31.14.137
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-3-129-26-249.us-east-2.compute.amazonaws.com open address
Hostname type -	Private IP DNS name (IPv4 only) -	

Connecting to EC2 Instance:

Connect to instance Info

Connect to your instance i-020a7650200bed52f (TECH Start) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID
i-020a7650200bed52f (TECH Start)

Connection Type

☒ Connect using EC2 Instance Connect
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☐ Connect using EC2 Instance Connect Endpoint
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IPv4 address
3.129.26.249

IPv6 address
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel | Connect

Creating S3 Bucket:

Buckets are containers for data stored in S3.

AWS Region
US East (Ohio) us-east-2

Bucket type	Info
-------------	------

- General purpose

General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

[Directory](#)

Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

tech-start-data-bucket

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

- **ACLs disabled (recommended)**

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☒ **Block public access to buckets and objects granted through *new* access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☒ **Block public access to buckets and objects granted through *any* access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☒ **Block public access to buckets and objects granted through *new* public bucket or access point policies**

aws

Search

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Amazon S3

Buckets

Create bucket

Default encryption

Info

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Info

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the [Storage](#) tab of the [Amazon S3 pricing page](#).

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

☒ Enable

Advanced settings

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel

Create bucket

AMAZON S3 Bucket created:

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Amazon S3

Amazon S3

General purpose buckets

Directory buckets

Table buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Account snapshot - updated every 24 hours

All AWS Regions

View Storage Lens dashboard

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets

Directory buckets

General purpose buckets (1)

Info

All AWS Regions

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

Name	AWS Region	IAM Access Analyzer	Creation date
tech-start-data-bucket	US East (Ohio) us-east-2	View analyzer for us-east-2	April 27, 2025, 00:40:24 (UTC-05:00)

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Amazon S3

Buckets

tech-start-data-bucket

tech-start-data-bucket

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Bucket overview

AWS Region
US East (Ohio) us-east-2

Amazon Resource Name (ARN)
[arn:aws:s3:::tech-start-data-bucket](#)

Creation date
April 27, 2025, 00:40:24 (UTC-05:00)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
Disabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Edit

Tags (0)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

Edit

Project Grading:

Your project will be graded based on how well you have performed the following:

- The effectiveness of your creation and assignment of various user accounts with differing access levels.
- The proficiency in your creation of an EC2 instance and the implementation of the S3 Storage option.
- The accuracy and completeness of your responses to the knowledge check questions.